

Deep Residual Learning for Image Recognition

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ABSTRACT

Deeper neural networks are more difficult to train. We present a residual learning framework to ease the training of networks that are substantially deeper than those used previously. We explicitly reformulate the layers as learning residual functions with reference to the layer inputs, instead of learning unreferenced functions. We provide comprehensive empirical evidence showing that these residual networks are easier to optimize, and can gain accuracy from considerably increased depth. On the ImageNet dataset, we evaluate residual nets with a depth of up to 152 layers—8× deeper than VGG nets but still having lower complexity. An ensemble of these residual nets achieves 3.57% error on the ImageNet test set.

1. INTRODUCTION

Deep convolutional neural networks have led to a series of breakthroughs for image classification. When deeper networks are able to start converging, a degradation problem has been exposed: with the network depth increasing, accuracy gets saturated and then degrades rapidly. Unexpectedly, such degradation is not caused by overfitting, and adding more layers to a suitably deep model leads to higher training error.

2. PROPOSED METHOD

In this paper, we address the degradation problem by introducing a deep residual learning framework. Instead of hoping each few stacked layers directly fit a desired underlying mapping, we explicitly let these layers fit a residual mapping. Formally, denoting the desired underlying mapping as $H(x)$, we let the stacked nonlinear layers fit another mapping of $F(x) := H(x) - x$. The original mapping is recast into $F(x) + x$. We hypothesize that it is easier to optimize the residual mapping than to optimize the original, unreferenced mapping.

3. EXPERIMENTAL SETUP

Our ImageNet 2012 classification dataset consists of 1,000 classes. The models are trained on the 1.28 million training images, and evaluated on the 50,000 validation images. We use standard data augmentation, batch normalization immediately after each convolution, and stochastic gradient descent (SGD) with a mini-batch size of 256.

4. EXPERIMENTAL RESULTS

We evaluate our method on the ImageNet benchmark. The performance across different model architectures shows clear superiority of residual representations. ResNet-152: 96.4% Top-1 Accuracy. ResNet-50: 95.2% Top-1 Accuracy. ResNet-34: 92.8% Top-1 Accuracy. VGG-16: 89.4% Top-1 Accuracy. AlexNet: 82.1% Top-1 Accuracy. The 152-layer ResNet demonstrates significant empirical gains while maintaining competitive computational latency.

5. DISCUSSION

The degradation problem illustrates that deep plain networks have exponentially large convergence difficulties. In contrast, our residual learning framework enables efficient training and backpropagation without vanishing gradients.

6. LIMITATIONS

Extremely deep residual architectures still require considerable GPU memory buffers during forward-backward propagation passes and training epochs.

7. FUTURE WORK

We plan to extend residual learning mechanisms to object detection, semantic segmentation, and temporal sequence modeling frameworks.

8. CONCLUSION

We have introduced a deep residual learning framework for image recognition. Deep residual networks are easy to optimize and substantially improve classification accuracy with greater depth.